

# Day 14 — Fri 18 Sep

**Countdown:** 8 days to General English & GK

**Method:** start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

## 🕒 GK — Mental Ability (all types)

Source: General/03\_GK\_CAPSULE.md

### 3.1 The road to statehood — the timeline

**Intuition.** The Naga political movement predates Indian independence. The path to statehood ran through decades of assertion of a distinct identity, culminating in a negotiated settlement — the 16-Point Agreement — that created the state in 1963.

| Year       | Event                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1918       | <b>Naga Club</b> formed (Kohima) — early expression of Naga political consciousness                                                         |
| 1929       | Naga Club submits a <b>memorandum to the Simon Commission</b> asking to be left out of the reformed India                                   |
| 1946       | <b>Naga National Council (NNC)</b> formed; <b>A. Z. Phizo</b> becomes a leading figure                                                      |
| 1957       | <b>Naga Hills–Tuensang Area (NHTA)</b> created (Naga Hills district of Assam + Tuensang) under the Centre                                   |
| 1960       | <b>16-Point Agreement</b> between the <b>Naga People's Convention (NPC)</b> and the Government of India — the political basis for statehood |
| 1962       | <b>State of Nagaland Act, 1962</b> passed; <b>13th Amendment</b> inserts Article 371A                                                       |
| 1 Dec 1963 | <b>Nagaland becomes the 16th state of India</b> — inaugurated by President S. Radhakrishnan △ verify the inaugurating dignitary             |
| 1963       | <b>P. Shilu Ao</b> — first Chief Minister; <b>Vishnu Sahay</b> — first Governor △ verify                                                    |
| 1975       | <b>Shillong Accord</b> (a controversial agreement with a section of the underground)                                                        |
| 1997       | <b>Ceasefire</b> between the Government of India and NSCN(IM) △ verify year                                                                 |
| 2015       | <b>Framework Agreement</b> signed towards a Naga political settlement △ verify                                                              |

**Memory chain:** 1918 Naga Club → 1929 Simon memorandum → 1946 NNC → 1960 16-Point Agreement → 1 Dec 1963 statehood. If you memorise one chain from this file, this is it.

**16-Point Agreement — what to know:** it was negotiated by the **Naga People's Convention** with the Government of India (1960) and led directly to the creation of Nagaland as a state under the Ministry of External Affairs initially, with Article 371A as its constitutional guarantee. △ verify the specific administrative point about MEA.

### 4.3 Modern India — the freedom-struggle timeline (learn cold)

**Intuition.** The exam loves dated events. Learn this as a single spine; hang everything else off it.

| Year | Event |
|------|-------|
|------|-------|

|         |                                                                                                                                      |
|---------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1757    | Battle of <b>Plassey</b> — British power begins in Bengal                                                                            |
| 1857    | <b>Revolt of 1857</b> ("First War of Independence"); Mangal Pandey; end of Mughal rule; Crown takes over (1858)                      |
| 1885    | <b>Indian National Congress</b> founded (A. O. Hume; first session Bombay, W. C. Bonnerjee)                                          |
| 1905    | <b>Partition of Bengal</b> (Curzon) → <b>Swadeshi Movement</b>                                                                       |
| 1906    | <b>Muslim League</b> founded (Dhaka)                                                                                                 |
| 1915    | <b>Gandhi returns</b> from South Africa                                                                                              |
| 1919    | <b>Jallianwala Bagh massacre</b> (Amritsar, 13 April); Rowlatt Act; Government of India Act 1919 (Montagu–Chelmsford)                |
| 1920–22 | <b>Non-Cooperation Movement</b> ; called off after <b>Chauri Chaura</b> (1922)                                                       |
| 1928    | Simon Commission protest; Lala Lajpat Rai injured                                                                                    |
| 1930    | <b>Dandi Salt March</b> (12 March) → <b>Civil Disobedience Movement</b> ; <b>Purna Swaraj</b> declared (26 Jan 1930, Lahore session) |
| 1931    | Gandhi–Irwin Pact                                                                                                                    |
| 1935    | <b>Government of India Act, 1935</b> (blueprint for much of the Constitution)                                                        |
| 1942    | <b>Quit India Movement</b> ("Do or Die"); INA under <b>Subhas Chandra Bose</b> ("Give me blood...")                                  |
| 1946    | Cabinet Mission; Constituent Assembly; Naval Mutiny                                                                                  |
| 1947    | <b>Independence, 15 August</b> ; Partition                                                                                           |

**Key figures — one line each:** **Gandhi** (non-violence, Satyagraha); **Nehru** (first PM); **Sardar Patel** ("Iron Man", integration of princely states); **Bhagat Singh** (revolutionary, executed 1931); **Subhas Chandra Bose** (INA, Azad Hind); **Ambedkar** (Constitution, social reform); **Tilak** ("Swaraj is my birthright").

## 8.1 Number & letter series

**Shortcut:** check the *difference* between consecutive terms; if constant → arithmetic; if multiplying → geometric; if differences themselves form a pattern → second-order.

**Worked:** 3, 6, 11, 18, 27, ? → differences 3,5,7,9 → next diff 11 → **27 + 11 = 38**.

### Practice (answers inside)

1. 2, 6, 12, 20, 30, ? → diffs 4,6,8,10 → **\*\*42\*\*** 2. 5, 10, 20, 40, ? →  $\times 2$  → **\*\*80\*\*** 3. 1, 4, 9, 16, 25, ? → squares → **\*\*36\*\*** 4. Letters: B, D, G, K, ? → +2,+3,+4 → **\*\*P\*\*** (K+5) 5. 7, 14, 28, 56, ? →  $\times 2$  → **\*\*112\*\***

## 8.2 Coding–decoding

**Shortcut:** find the shift between letters (position in alphabet). A=1 ... Z=26.

**Worked:** if CAT = DBU, each letter +1 → so DOG = **EPH**.

### Practice

1. If MANGO is coded as NBOHP, then APPLE = ? → each +1 → **\*\*BQQMF\*\*** 2. If 24 = "CX" pattern... \*(stick to letter-shift types in this paper)\* 3. If SUN = TVO (each +1), MOON = ? → **\*\*NPPO\*\***

## 8.3 Blood relations

**Shortcut:** draw a family tree; use + for male, – for female; work from the speaker outward. "**Only son of my mother's father**" = my maternal grandfather's only son = **my maternal uncle** (or father, if worded so — read carefully).

**Worked:** "Pointing to a man, a woman said, 'His mother is the only daughter of my mother.'" → the only daughter of the woman's mother = the woman herself → so the man's mother = the woman → **the man is her son.**

#### Practice

1. "A is B's sister. C is B's mother. D is C's father. How is A related to D?" → A is C's daughter → C is D's daughter → **\*\*A is D's granddaughter.\*\*** 2. "If P is the brother of Q, and Q is the sister of R, how is P related to R?" → **\*\*P is R's brother.\*\*** (assuming P is male)

## 8.4 Direction sense

**Shortcut:** draw N-up, E-right axes; track each move; net displacement by Pythagoras if asked for distance. Remember at sunrise your shadow falls **west** (sun in east).

**Worked:** walk 3 km North, then 4 km East. Distance from start =  $\sqrt{3^2 + 4^2} = 5 \text{ km}$ ; direction = **North-East**.

#### Practice

1. Walk 5 km South, turn left, walk 5 km. Which direction do you now face? → started facing S, **\*\*turn left** → East. **\*\*** (Left of South is East.) 2. 6 km East then 8 km North → straight-line distance =  $\sqrt{36+64} = 10 \text{ km}$ . **\*\***

## 8.5 Syllogism

**Shortcut:** use Venn diagrams. "All A are B" = A circle inside B. A conclusion is valid only if it is true in **every** possible diagram.

**Worked:** *All cats are animals. All animals are living.* → **All cats are living** (valid); "Some living are cats" (valid). "All animals are cats" (invalid).

#### Practice

1. \*All pens are books. Some books are red.\* → "Some pens are red"? → **\*\*Invalid\*\*** (not necessarily). "Some books are pens"? → **\*\*Valid\*\*** (conversion of "all pens are books"). 2. \*No dog is a cat. All cats are pets.\* → "No dog is a pet"? → **\*\*Invalid.\*\*** "Some pets are not dogs"? → **\*\*Valid.\*\***

## 8.6 Analogy

**Shortcut:** name the *relationship* in the first pair, then apply it.

**Worked:** Doctor : Hospital :: Teacher : ? → workplace → **School.**

#### Practice

1. Hand : Glove :: Foot : ? → **\*\*Sock/Shoe\*\*** 2. Cub : Bear :: Kitten : ? → **\*\*Cat\*\*** 3. India : Rupee :: Japan : ? → **\*\*Yen\*\***

## 8.7 Ratio & proportion

**Shortcut:** a ratio a:b can be scaled; if a:b = c:d then **ad = bc** (cross-multiply).

**Worked:** divide ₹600 between A and B in ratio 2:3 → total parts 5 → one part = 120 → **A = ₹240, B = ₹360.**

#### Practice

1. Divide 750 in ratio 3:2 → parts 5, one part 150 → **\*\*450 and 300.\*\*** 2. If 4:5 = x:20 →  $x = (4 \times 20)/5 = 16$ . **\*\***

## 8.8 Percentages

**Shortcut:**  $x\%$  of  $y = (x/100) \cdot y = \mathbf{x \cdot y / 100}$ . To increase by  $x\%$ , multiply by  $(1 + x/100)$ . Percentage change =  $(\text{change}/\text{original}) \times 100$ .

**Worked:** a shirt costs ₹800, increased 15%  $\rightarrow 800 \times 1.15 = \mathbf{₹920}$ .

### Practice

1. 25% of 480 = **120**. 2. A number increased from 200 to 250  $\rightarrow$  % increase =  $(50/200) \times 100 = \mathbf{25\%}$ . 3. If price falls 20% from ₹500  $\rightarrow 500 \times 0.8 = \mathbf{₹400}$ .

## 8.9 Profit & loss

**Shortcut:** **Profit%** = **(Profit / Cost Price)  $\times$  100** (always on CP unless stated).

$SP = CP \times (1 + \text{profit}\%/100)$ .

**Worked:** CP ₹400, SP ₹500  $\rightarrow$  profit ₹100  $\rightarrow$  profit% =  $100/400 \times 100 = \mathbf{25\%}$ .

### Practice

1. CP ₹250, SP ₹200  $\rightarrow$  loss = 50  $\rightarrow$  loss% =  $50/250 \times 100 = \mathbf{20\%}$ . 2. CP ₹1000, profit 12%  $\rightarrow$  SP =  $1000 \times 1.12 = \mathbf{₹1120}$ .

## 8.10 Time & work

**Shortcut:** if A does a job in  $n$  days, A's one-day work =  $\mathbf{1/n}$ . Add rates for people working together.

**Worked:** A alone 10 days, B alone 15 days  $\rightarrow$  together  $1/10 + 1/15 = 3/30 + 2/30 = 5/30 = 1/6 \rightarrow \mathbf{6 \text{ days}}$ .

### Practice

1. A: 12 days, B: 12 days  $\rightarrow$  together =  $1/12 + 1/12 = 1/6 \rightarrow \mathbf{6 \text{ days}}$ . 2. A does a job in 20 days; works 5 days then leaves — fraction done =  $5/20 = \mathbf{1/4}$ ;  $\mathbf{3/4}$  remains.

## 8.11 Time, speed & distance

**Shortcut:** **Speed = Distance / Time**. To convert km/h  $\rightarrow$  m/s multiply by  $\mathbf{5/18}$ ; m/s  $\rightarrow$  km/h multiply by  $\mathbf{18/5}$ .

**Worked:** 72 km/h in m/s =  $72 \times 5/18 = \mathbf{20 \text{ m/s}}$ .

### Practice

1. A car travels 150 km in 3 h  $\rightarrow$  speed =  $\mathbf{50 \text{ km/h}}$ . 2. 36 km/h  $\rightarrow$  m/s =  $36 \times 5/18 = \mathbf{10 \text{ m/s}}$ . 3. A train 100 m long at 20 m/s crosses a pole in  $\rightarrow 100/20 = \mathbf{5 \text{ s}}$ .

## 8.12 Averages

**Shortcut:** **Average = sum of terms / number of terms**. Average of first  $n$  natural numbers =  $\mathbf{(n+1)/2}$ .

**Worked:** average of 10, 20, 30, 40 =  $100/4 = \mathbf{25}$ .

### Practice

1. Average of 5, 15, 25 =  $45/3 = \mathbf{15}$ . 2. Average of first 10 natural numbers =  $(10+1)/2 = \mathbf{5.5}$ .

## 8.13 Simple & compound interest

**revision routine** and **reliable sources**. Fill it in yourself from real sources in the run-up to 26 September.

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## Core — disk scheduling & RAID

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Study **\_Shared/CORE\_03\_OperatingSystems.md** today (section match not found — open the full topic PDF for: disk, raid, file system, protection, security).